Case 02 · Ideation document: Maturity Wall Simulator

Following chapter 1 (Initiation) of the Magshimim project-book template. For each project: overview, goal, target audience, existing solutions, technological core, technologies, challenges and sources, hardware requirements, and conclusion.

Overview

A tool that shows, for any country and bond by bond, when its cheap debt matures and at what rate it will be replaced. It does this by collecting the public maturity schedules published by finance ministries and running yield scenarios over every bond, so that the "maturity wall" becomes something you see years ahead rather than something you discover in the budget after the fact.

In practice: the user picks a country (say France) and sees all its debt as a "wall" of bonds stacked by maturity year. Moving a slider for future yields changes the wall's colours: every bond that will be refinanced above its original coupon is coloured by the size of the jump. Above the wall runs a debt engine that computes the debt-to-GDP ratio year by year and marks the moment when r overtakes g and debt starts to grow by itself.

Project goal

We chose this project because the story reveals something odd: information about the maturity wall is entirely public. Every finance ministry publishes which bonds mature and when. Yet to estimate what refinancing would cost France, The Economist had to run a thought experiment of its own ("suppose all the debt were rolled over today"). The data exist, but there is no open tool that turns them into a forecast.

The goal: close the gap between "the information exists" and "someone actually sees it". An open, transparent, reproducible tool in which anyone can check the calculation, and which shows not only how much the debt will cost but also when and how fast.

Target audience

- **Primary audience:** economics journalists, analysts at think-tanks and parliamentary aides, with basic economics knowledge. They need a reliable, explained number quickly, without building a model themselves.
- **Secondary audience:** economics and computer-science students, and citizens who want to understand what the story means. No prior knowledge is required: every term is explained inside the tool.
- **Illustrative scenario:** an economics reporter in Paris receives Mr Lecornu's draft budget in the morning. Within ten minutes she knows how much of the €43bn will be swallowed by debt refinancing alone in 2027.

Existing solutions

- **IMF DSA framework (Debt Sustainability Analysis):** the IMF runs debt-sustainability analyses for every country, including a public spreadsheet template (MAC SRDSF). Very strong, but it works on aggregate debt rather than bond by bond, and is updated once a year as part of the country report.
- **CBO and CRFB in America:** publish long-term projections of interest payments (the source of the "$2.8trn" in the story). Reliable, but only for America, and as a report rather than a tool you can play with.
- **Treasury Fiscal Data and Agence France Trésor:** publish the full list of bonds with maturity dates and coupons. This is our raw material, not a forecast.
- **Bloomberg and trading terminals:** show "maturity walls" to subscribers. Not open, not transparent in the calculation, and not connected to an r versus g engine.
- **Think-tanks (Bruegel, for example):** have published open DSA models for European countries. Close in spirit, but research tools, not a product that updates itself.

What sets us apart: the combination of bond-level detail, interactive scenarios and an r versus g engine, in an open tool that updates automatically from official sources. The novelty is in the combination and the accessibility, not in the mathematics itself, and that is a question mark we must carry along the way.

Technological core

The core is the rollover engine: an algorithm that takes a list of bonds (amount, coupon, maturity date) and a future yield curve, and computes year by year which debt matures, at what rate it is replaced, and the cumulative interest cost. On top of it sits the stock-flow debt-dynamics equation: next year's debt equals this year's debt times (1+r)/(1+g), minus the primary surplus.

Without this engine there is no project: the visualisation, country comparisons and alerts are just ways of presenting its output.

Main technologies

- **Python with Polars:** to process tables of thousands of bonds quickly. We chose Polars over pandas for its columnar performance and more predictable type behaviour.
- **Official APIs:** Treasury Fiscal Data (America), the ECB Data Portal (European yields and curves), and the publication files of debt offices (France, Italy). We use official sources only, so every number can be traced.
- **DuckDB:** an in-file analytical database that keeps a version of every load. This lets us ask "what did we see a month ago?" and compare.
- **FastAPI:** a thin API layer serving the engine's results.
- **D3.js:** for the interactive maturity wall.

Open questions:

- **Deterministic scenarios versus Monte Carlo simulation:** fixed scenarios (yields stay / rise 1% / fall) are easy to explain. Monte Carlo gives a distribution and probabilities, but is harder

for a reader without background. We will probably start with scenarios and check during the POC whether a distribution adds anything.

- How to handle inflation-linked bonds and floating-rate debt, which behave differently from ordinary bonds.

Technological challenges and sources

- **Inconsistent data:** every debt office publishes in a different format (CSV, Excel, PDF, API). We will build an adapter per source, with a reconciliation check: the sum of all bonds must match the marketable debt the country reports. Sources: fiscaldata.treasury.gov, aft.gouv.fr, data.ecb.europa.eu.
- **The rollover assumption:** the engine assumes each maturing bond is replaced by a bond of the same tenor. In reality debt offices change their mix of maturities. The assumption must be shown openly and be adjustable.
- **Debt that is not in the lists:** local-government debt, state guarantees. We will define explicitly what is in scope and what is not.
- **Validation:** how do we know the engine is right? We will reproduce a known projection (CBO for America, The Economist's exercise for France) and check that we land in the same order of magnitude. Source: the IMF DSA methodology documents.

Hardware requirements

None. The engine runs on an ordinary computer or a small cloud server. All the bonds of several countries come to tens of thousands of rows at most.

Conclusion

✅ Survives, with a condition. Public data, checkable mathematics, and a fully real POC is possible. The condition: novelty. Official tools already do much of this, so in the research stage we must show that the combination (bond level, scenarios, r versus g, open) yields an insight the existing tools do not.